Biomimetic Intelligence and Robotics 3 (2023) 100100

Contents lists available at ScienceDirect

Biomimetic Intelligence and Robotics

journal homepage: www.elsevier.com/locate/birob

Research

An online impedance adaptation controller for decoding skill intelligence

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ARTICLE INFO

Article history:
Received 31 October 2022
Revised 22 February 2023
Accepted 28 February 2023
Available online 13 March 2023

Keywords:
Variable impedance control
Wearable robots
Motor control
Deformable object manipulation

ABSTRACT

Variable Impedance control allows robots and humans to safely and efficiently interact with unknown external environments. This tutorial introduces online impedance adaptation control (OIAC) for variable compliant joint motions in a range of control tasks: rapid (< 1 *s*) movement control (i.e., whipping to hit), arm and finger impedance quantification, multifunctional exoskeleton control, and robot-inspired human arm control hypothesis. The OIAC has been introduced as a feedback control, which can be integrated into a feedforward control, e.g., learned by data-driven methods. This integration facilitates the understanding of human and robot arm control, closing a research loop between biomechanics and robotics. It shows not only a research way from biomechanics to robotics, but also another reserved one. This tutorial aims at presenting research examples and Python codes for advancing the understanding of variable impedance adaptation in human and robot motor control. It contributes to the state-of-the-art by providing an online impedance adaptation controller for wearable robots (i.e., exoskeletons) which can be used in robotic and biomechanical applications.

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1. Introduction

Mechanical impedance is an important measure of the relation between a force/torque acting on a mechanical system and the resulting deformation or motion [1]. Human and robot joint impedances are important when they physically interact with the external environment [2]. How they modulate their joint impedance in various interactive control tasks [3], is a key to successful execution of tasks, and therefore a popular topic in biomechanics and robotics communities.

This tutorial contributes to the state-of-the-art by providing applications and open-sourcing their implementation codes on online impedance adaptation control (OIAC) as a feedback controller (see Fig. 1), in which stiffness and damping parameters are online tuned in various applications. The tutorial is limited to a software and simple way for achieving variable compliant robot control. It was applied to three wearable robot applications, i.e., mechanical impedance quantification, assist- and resist-as-needed control. Together with feedforward control, the OIAC provides a simple solution to achieve variable impedance adaptations in a range of human and robot control, compared to existing methods on either complex learning mechanisms or a single scenario [4–6]. A review of state of the art can be seen at Section 2. Besides, the proposed physics-based model (i.e., OIAC) is integrated into feedforward counterparts such as a Gaussian neural network, showing that this integration facilitates the understanding of human and robot arm control (see Fig. 1(a)). For instance, one of its applications shows how a neural network leverages joint mechanics to simplify robot arm control, raising a leading joint hypothesis for future human validations. Furthermore, its applications close a research loop, i.e., from humans (biomechanics) to robots and back (see Fig. 1(b)). Specifically, it was inspired from human impedance control experiments [7], being transferred to bio-inspired control in human-like rapid (< 1 *s*) manipulation of a flexible object (see Section 4.1) and assisting (back to) human arm control by providing impedance quantification, multifunctional exoskeleton control, and a robot-inspired method.

The remainder of this tutorial is structured as follows. The following section includes a review of state of the art works. The OIAC is provided in Section 3. Section 4 contains four human and robot arm applications governed by the OIAC and feedforward control. A conclusion is concluded in Section 5.

2. State of the art (SOTA)

There is a trend to close and form a research loop between robots and humans (biomechanics), in which roboticists take

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https://doi.org/10.1016/j.birob.2023.100100
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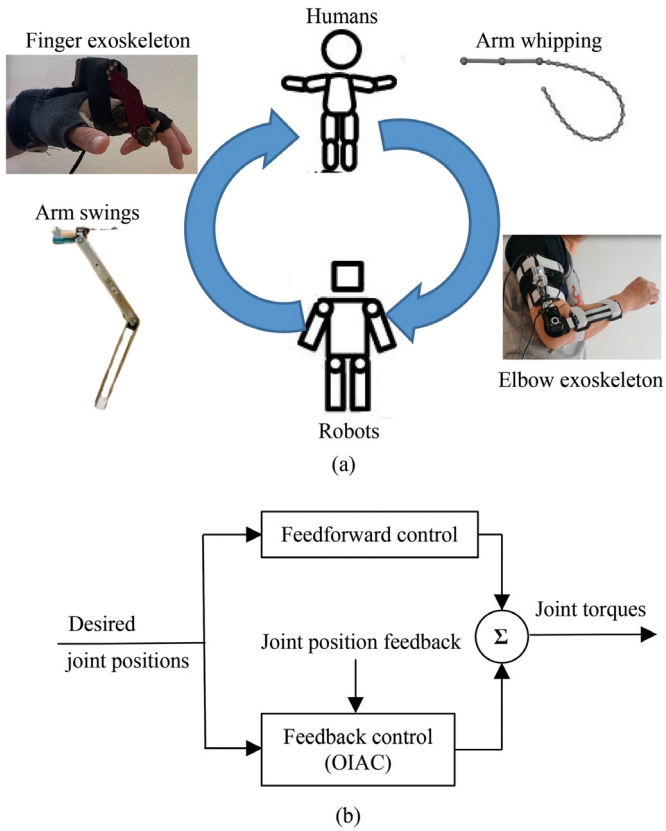

Fig. 1. Closing a research loop between humans (biomechanics) and robots to decode motion intelligence. (a) Exoskeletons and human-like movements control. (b) Feedforward and feedback control.

advantages of human motor control principles to build friendly robots while biomechanists can test hypotheses (on human motor control) that can be provided by robots and their controllers (as abstract models) [8]. The main contribution of the presented OIAC to the SOTA is to provide a simple method to decode a set of intelligent skills of humans and robots.

2.1. From biomechanics to robotics

It is manifested that humans as very flexible and efficient biological systems are able to regulate the stiffness of their hands by con-contracting the arm muscles when physically interacting with external environments, especially the environments with unknown mechanical properties [7]. For instance, increasing the hand stiffness to stabilize any possibly unstable physical interactions, or decreasing the stiffness to avoid any adverse large contact forces with unknown environments. This intelligence of the stiffness regulation exhibited by humans is very important for advancing our current robots to next generation of robots with high physical intelligence when performing tasks which require physical interactions with environments. Inspired by this human intelligence, researchers developed a human-like controller which can adapt the impedance and force in stable and unstable interactions [6]. In addition to the human hand stiffness, a method of estimating human arm joint stiffness was proposed and developed based on a human arm musculoskeletal model [9], and the estimated joint stiffness was used and transferred to the robot counterpart to conduct, for example, teleoperation tasks [10]. Recently, human joint stiffness estimates were also utilized as a reference and clue to adjust the joint stiffness of an arm exoskeleton to improve task performance and reduce energy consumption when operators carry out tasks by wearing the exoskeleton [11].

Regarding the human arm impedance estimation and transfer to robots, the current challenges lie in how to generalize the impedance adaptation strategy learnt from one task to novel tasks, and how to encode a flexible skill including both motion information and impedance adaptation, which can applied to different tasks [12]. One possible solution is to characterize the tasks' requirements in terms of motion and impedance parameters and parameterize a skill based on these parameters. Alternatively, reinforcement learning can be used to learn optimal impedance parameters during the interactions with the environments in new scenarios.

2.2. From robotics to biomechanics

Compared to the path from biomechanics to robotics, its reversed one on physical intelligence has not yet fully explored owing to less understanding of complex human motor control systems. This exploration can date back to paper [13], in which a 30-DOF hydraulic humanoid robot and learning algorithms were developed to reproduce and investigate human motions. However, it does not decode feedforward and feedback control principles in a range of physical interaction control tasks presented in this tutorial. This case happens in other studies. For instance, Nah et al. proposed a dynamic primitive model for uncovering control principles of rapid and complex movement control [14], i.e., whipping-to-hit. However, its feedback mechanism is a simple (constant) impedance control that does not afford to explain control principles in real human whipping, e.g., online impedance adaptation. Ruit et al. presented a serial of data-driven methods (e.g., kernel-based regression) to quantify human joint impedance [15], but they fail to capture nonlinear properties of online physics interaction. In addition to software

Table 1
Integration of feedback OIAC into feedforward control in human and robot arm control (see Fig. 1(a)). Symbol "-" represents no need for feedforward control.

Feedback	Feedforward	Application	Direction
OIAC	Offline optimized motion primitives	Whipping to hit	human ⟶ robot
OIAC	–	Quantify joint impedance	robot ⟶ human
OIAC	Iterative learning	Multifunctional exoskeleton control	human ⟶ robot
OIAC	Gaussian neural network	Arm swings	robot ⟶ human

methods, Chang et al. developed a hardware (exoskeleton) device for estimating average and static shoulder impedance [16]. However, the estimation was based on linear perturbations (resistances), which does not uncover online impedance adaptation to nonlinear resistances.

It is obvious that humans adapt their shoulder and elbow joint impedance to perform daily movements [1], e.g., arm swings when walking. However, it remains unknown whether there is a control gradient between shoulders and elbows. Specifically, the joints (e.g., shoulders) closed to main bodies are actively controlled, while those (e.g., elbows) far away move along with the closed, i.e., being (mainly) passively actuated. This is so called proximo-distal control gradient that had been validated in animals legs [17], but has not yet been found in human arms. To address it, we provided a robot-inspired method with online impedance adaptation control to propose a proximo-distal control hypothesis for biomechanists (see details in Section 4.4).

3. Online impedance adaptation control (OIAC)

The OIAC is a feedback-based PD controller for producing the joint torques $\boldsymbol{\tau}_{fb}$ of an n-link robot given by (see Fig. 1(b)),

$$\boldsymbol{\tau}_{fb} = \boldsymbol{K}(t)\,\boldsymbol{e}(t) + \boldsymbol{B}(t)\,\dot{\boldsymbol{e}}(t),\ \boldsymbol{e}(t) = \boldsymbol{q}_d(t) - \boldsymbol{q}(t),\ \dot{\boldsymbol{e}}(t) = (\dot{\boldsymbol{q}}(t)_d - \dot{\boldsymbol{q}}(t)) \tag{1}$$

where $\boldsymbol{K}(t)$ and $\boldsymbol{B}(t) \in \mathbb{R}^{n\times n}$ are the time-varying impedance (stiffness and damping) matrices. $\boldsymbol{q}(t)$ and $\boldsymbol{q}_d(t) \in \mathbb{R}^n$ the real and desired joint positions as well as their error $\boldsymbol{e}(t)$. $\dot{\boldsymbol{q}}_d$ and $\dot{\boldsymbol{q}} \in \mathbb{R}^n$ the real and desired joint velocities as well as their error $\dot{\boldsymbol{e}}(t)$.

The adaptation law of the impedance matrices $\boldsymbol{K}(t)$ and $\boldsymbol{B}(t)$ minimized the following scalar cost function J consisting of control effort J_c and error J_p [6],

$$J = J_c + J_p \tag{2}$$

The cost function J_c was defined as follows [6],

$$J_c = \frac{1}{2}\int_0^D \left(\left\| vec(\tilde{\boldsymbol{K}}) \right\|^2_{\boldsymbol{Q_K}} + \left\| vec(\tilde{\boldsymbol{B}}) \right\|^2_{\boldsymbol{Q_B}} \right) d\tau \tag{3}$$

which is the time integration of the weighted norms of matrices $\tilde{\boldsymbol{K}},\ \tilde{\boldsymbol{B}} \in \mathbb{R}^{n\times n}$; $\|\cdot\|_{\boldsymbol{Q_K},\boldsymbol{Q_B}}$ and $vec(\cdot)$ denote the weighted norms and column vectorization, respectively; $\boldsymbol{Q_K} \in \mathbb{R}^{n\times n}$ and $\boldsymbol{Q_B} \in \mathbb{R}^{n\times n}$ are positive symmetric weighting matrices, where the norm of the matrix affects the speed of optimization [18]; $\tilde{\boldsymbol{K}}$ and $\tilde{\boldsymbol{B}}$ are defined as the difference between the actual and initial values of the stiffness and damping matrices [18],

$$\tilde{\boldsymbol{K}} = \boldsymbol{K} - \boldsymbol{K}_n,\quad \tilde{\boldsymbol{B}} = \boldsymbol{B} - \boldsymbol{B}_n \tag{4}$$

where subscript $\boldsymbol{K}_n$ and $\boldsymbol{B}_n$ denote initial impedance (matrices) to ensure movement stability, from which the optimal $\boldsymbol{K}$ and $\boldsymbol{B}$ are optimized.

The cost function J_p is defined as follows [6],

$$J_p = \int_0^D \dot{V}(\tau)d\tau,\ V(t) = \frac{1}{2}\boldsymbol{\varepsilon}^T(t)\boldsymbol{I}(\boldsymbol{q})\boldsymbol{\varepsilon}(t) \tag{5}$$

where the integrand is the time derivative of the scalar function $V(t)$; $\boldsymbol{I}(\boldsymbol{q}) \in \mathbb{R}^{n\times n}$ is the inertia matrix; $\boldsymbol{\varepsilon}(t)$ is the time-varying sliding variable [19], defined as follows,

$$\boldsymbol{\varepsilon}(t) = \boldsymbol{e}(t) + \beta\dot{\boldsymbol{e}}(t),\ \boldsymbol{e}(t) = \boldsymbol{q}_d(t) - \boldsymbol{q}(t) \tag{6}$$

Table 2
Parameters of whip models in Mojoco [22] (see Fig. 3 (a)).

Type	Nodes N	Mass m	Length l	Stiffness k	Damping b
short	10	0.1	0.1	0.05	0.005
medium	15	0.1	0.1	0.05	0.005
long	25	0.012	0.072	0.242	0.092

where $\boldsymbol{e}(t) \in \mathbb{R}^n$ is the tracking error between the zero-torque trajectory and the actual joint trajectory of the upper-limb model; β is a positive constant.

The impedance matrices $\boldsymbol{K}(t)$ and $\boldsymbol{B}(t)$ are online tuned by the following rule [18],

$$\boldsymbol{K}(t) = \boldsymbol{F}(t)\boldsymbol{e}^T(t),\ \boldsymbol{B}(t) = \boldsymbol{F}(t)\dot{\boldsymbol{e}}^T(t),\ \boldsymbol{F}(t) = \frac{\boldsymbol{\varepsilon}(t)}{\gamma(t)} \tag{7}$$

where $\gamma(t)$ is an adaptation scalar defined as:

$$\gamma(t) = \frac{a}{1 + b\,\|\boldsymbol{\varepsilon}(t)\|^2},\ \boldsymbol{\varepsilon}(t) = \boldsymbol{e}(t) + \beta\dot{\boldsymbol{e}}(t) \tag{8}$$

The values of the positive scalars are set as $a = 0.2$, $b = 5$ and $\beta = 0.05$. This adaptation law was derived by minimizing control effort and error. Further details and derivation of the online impedance adaptation law (Eq. (7)) were presented in [18].

4. Human and robot arm control applications

The OIAC as a feedback based controller, was applied to the four applications human-like rapid ($< 1\ s$) manipulation of a flexible object, mechanical (interactive) impedance quantification of human fingers and elbows, multifunctional exoskeleton control, and a robot-inspired human arm control hypothesis. In three of them, it was integrated into data-driven methods such as a Gaussian neural network (see Table 1). A list of questions were answered in the four applications.

- How does online impedance adaptation facilitate arm control in manipulating a flexible object (i.e., whip)?
- How can an exoskeleton and its controller be used to quantify mechanical impedance of human fingers and elbows?
- How can an exoskeleton be multifunctional for fitness and rehabilitation applications?
- How do neural mechanisms leverage joint mechanics to simplify arm movement control? how can a hypothesis of human arm control be raised by using its robot counterpart?

4.1. Human-like rapid manipulation of a flexible object

The artists can manipulate whips to hit targets (e.g., bottle caps). How to compute and simulate these fascinating arm manipulation control is a challenging question, since it has to handle rapid ($< 1\ s$) movements and (theoretically) infinite degrees of freedom of a flexible object [20]. To address it, Hogan and Sternad proposed a dynamic primitive movement framework consisting of interactive (impedance) and motion primitives [21] (see Fig. 2). However, how time-varying impedance control contributes to this arm manipulation control remains unclear.

In our work [23], a two jointed arm attached with a whip was simulated in the physics simulator Mujoco [22]. Each whip was

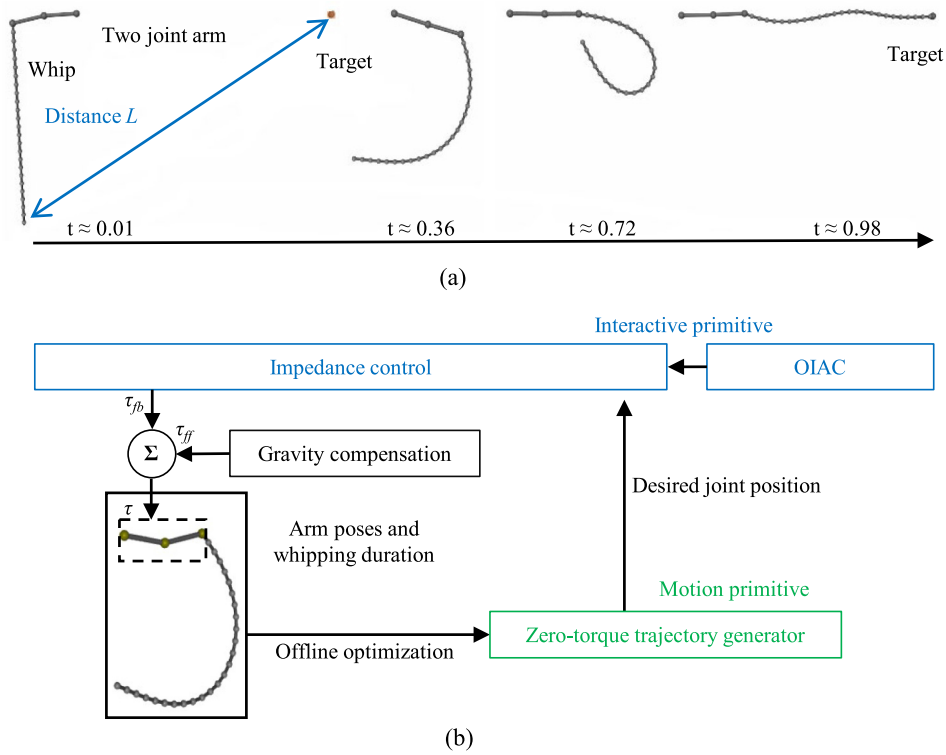

Fig. 2. Integration of the OIAC into dynamics motor primitives (DMPs) for arm control when whipping to hit a target. (a) Time sequence of whipping-to-hit. (b) Integrated control framework.

modeled as a chained rotational mass–spring–damper system with N nodes. Each sub-part of a whip consists of a point-mass m [kg] and a rotational joint suspended with a massless cylinder with length l [m]. The stiffness and damper parameters rotational joints denote k [N m/rad] and b [N m s/rad] (see Table 2). All listed parameters were derived from real whip movements [14]. The whip was manipulated to hit a target by (torque) controlling two arm joints. Following by a dynamic motor primitive framework, the controller consists of (feedforward) gravity compensation, motion (planning) and interactive (impedance) primitives,

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{ff} + \boldsymbol{\tau}_{fb} \tag{9}$$

where $\boldsymbol{\tau}_{ff} \in \mathbb{R}^{2\times 1}$ is the feedforward joint torque computed by gravity compensation,

$$\boldsymbol{\tau}_{ff} = \boldsymbol{J}_1^T \boldsymbol{f}_1 + \boldsymbol{J}_2^T \boldsymbol{f}_2 + \boldsymbol{J}_3^T \boldsymbol{f}_3 \tag{10}$$

where $\boldsymbol{J}_{1-3} \in \mathbb{R}^{3\times 2}$ are the Jacobian matrices in terms of the center of mass of the upper arm, center of mass of the forearm, and the end-effector with respect to the frame attached to the shoulder, respectively. $\boldsymbol{f}_{1-3} \in \mathbb{R}^3$ denote the gravitational forces due to the mass of the upper arm, forearm, and whip model, respectively [14],

$$\boldsymbol{f}_1 = M_1 \boldsymbol{g}, \quad \boldsymbol{f}_1 = M_2 \boldsymbol{g}, \quad \boldsymbol{f}_3 = M_w \boldsymbol{g} \tag{11}$$

where $M_1 = 0.291$ kg and $M_2 = 0.294$ kg denote the mass of the upper arm and forearm, respectively [14]; M_w denotes the total mass of the whip model, which is equal to the node number of the whip N times the mass of each sub-model m ($M_w =$ N m); $\boldsymbol{g} \in \mathbb{R}^3$ denotes gravity in the simulation environment. In addition, the OIAC was introduced into the interactive primitive for online modulating stiffness and damper parameters of impedance control (see Fig. 2), Specifically, $\boldsymbol{\tau}_{fb} \in \mathbb{R}^{2\times 1}$ in Eq. (9) is the feedback joint torque computed by Eqs. (1) and (7). Its required desired joint angles $\boldsymbol{q}_d(t)$ are given by a minimum jerk trajectory, a minimum-jerk trajectory [14]:

$$\boldsymbol{q}_d(t) = \boldsymbol{q}_i + (\boldsymbol{q}_f - \boldsymbol{q}_i)\left\{10\left(\frac{t}{D}\right)^3 - 15\left(\frac{t}{D}\right)^4 + 6\left(\frac{t}{D}\right)^5\right\} \tag{12}$$

where $\boldsymbol{q}_i = [q_{i,s}, q_{i,e}]^T$ $[rad]$ and $\boldsymbol{q}_f = [q_{f,s}, q_{f,e}]^T$ $[rad]$ denote the initial and final zero-torque joint posture, respectively; D $[s]$ is the duration of the trajectory; subscripts s and e denote shoulder and elbow joints, respectively. For times longer than the duration D (i.e. $t > D$), the zero-torque trajectory $\boldsymbol{q}_d(t)$ remained at $\boldsymbol{q}_f$ [14]. The trajectory parameters $(q_{i,s}, q_{i,e}, q_{f,s}, q_{f,e}, D)$ were offile optimized by the "Dividing rectangles-locally biased" (DIRECT-L) algorithm in the "nonlinear optimization Python tool box" (see Algorithm 1). The whip and control model link can be seen at footnote.[1]

The introduction of the OIAC into the DMPs was compared with constant impedance control given by,

$$\boldsymbol{K} = \begin{bmatrix} 29.5 & 14.3 \\ 14.3 & 39.3 \end{bmatrix}, \quad \boldsymbol{B} = 0.1\boldsymbol{K} = \begin{bmatrix} 2.95 & 1.43 \\ 1.43 & 3.93 \end{bmatrix} \tag{13}$$

It was hypothesized that this introduction facilitates the whip-to-hit movement control over constant impedance control. Specifically, the OIAC introduction allows for more accurate tracking of planned joint motions, compared to constant impedance control [14] (see Fig. 3(b)). As a result, it (1) accelerates offline motion (primitive) optimization (see Fig. 3(c)), (2) results in lower stiffness and high damping averages in variable impedance control (see Fig. 3(d)). These results facilitate our understanding of feedback-based impedance control in rapidly manipulating a flexible object. The experimental video link can be seen at footnote.[2]

[1] https://github.com/mosesnah-shared/whip-project-targeting
[2] https://www.youtube.com/watch?v=AV_7qP6Yd-Y

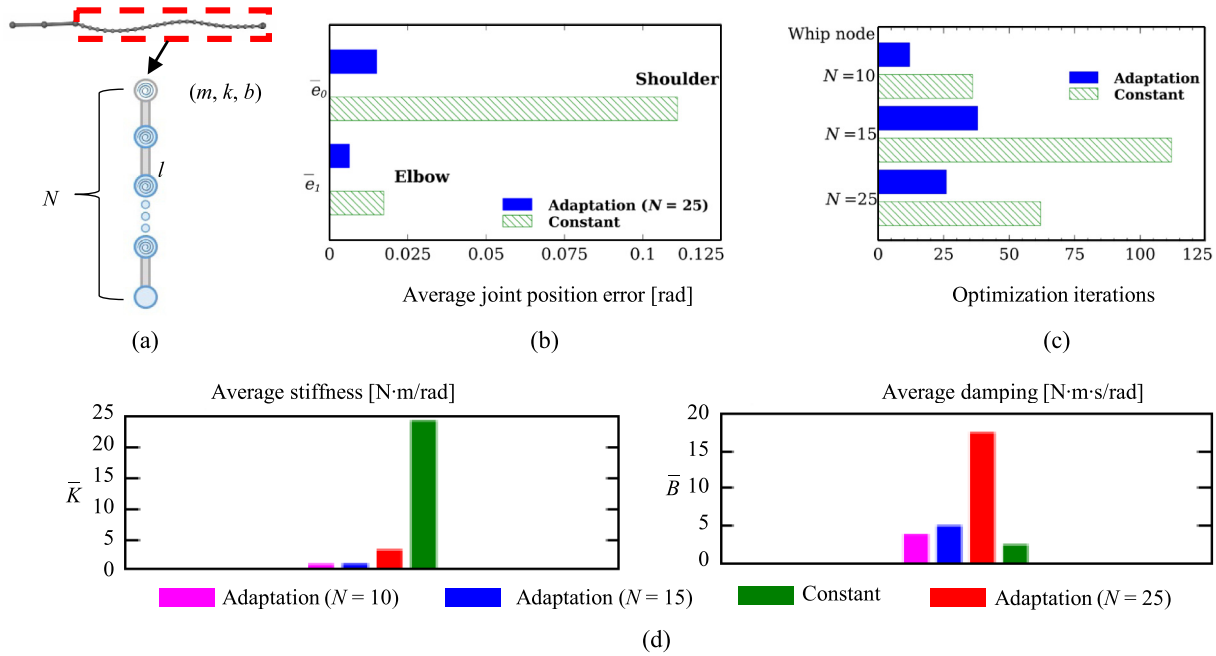

Fig. 3. Whip-to-hit movement control. (a) Whip models (see parameters in Table 2). (b) Average shoulder and elbow joint error. (c) Offline optimization iterations of zero torque trajectory parameters (see Eq. (12)). (d) Average stiffness and damping.

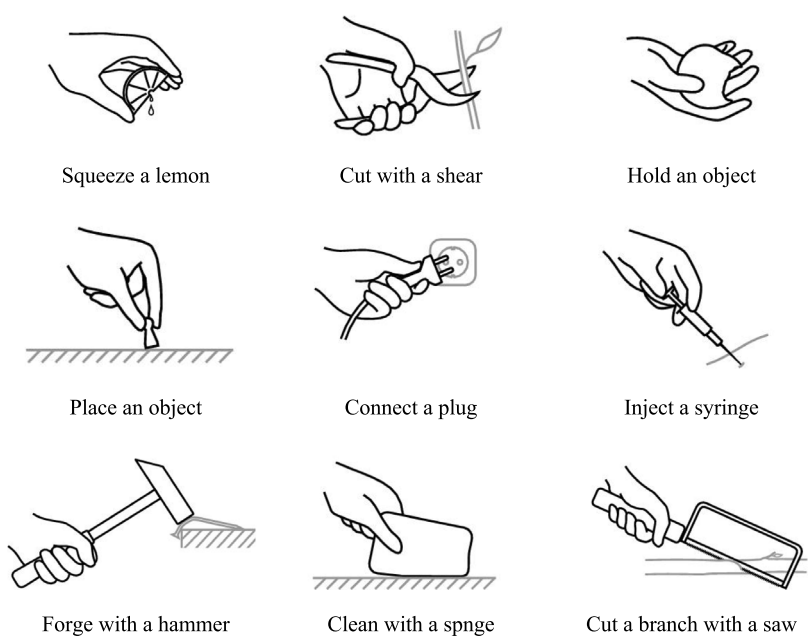

Fig. 4. A range of compliant object manipulation in daily life [24].

4.2. Mechanical impedance quantification

Humans can effortlessly manipulate a range of variable compliant objects in daily life (see Fig. 4), e.g., squeezing lemons. During the manipulations, physical resistances nonlinearly change owing to intrinsic object compliance. It goes without saying that humans need to tune mechanical joint impedance to physically manipulate these objects [1]. However, decoding how humans adapt its joint impedance to manipulated object compliance remained a challenging problem. This is because there was no mechanism developed for simulating nonlinear (physical) resistance of manipulated objects, rather heavy multi-joint (e.g., Kuka) robot arms and force sensors were used to estimate human arm impedance, e.g., in [25]. This reduces their wide applications and experimental costs. Therefore, how to quantify human arm impedance in a portable and cost-effective was an open question. To address it, the OIAC has been applied to portable finger (TIE-EXO, 0.25 kg) and elbow (POW-EXO, 0.425 kg) exoskeletons, serving as a 'sensor' for quantifying mechanical impedance in nonlinear resistance [26,27] (see Fig. 5). The TIE-EXO consists of a soft glove, Dynamixel actuator (XM430-W350-R), rigid shaft, USB communication interface, power converter, and three soft strip bands. The POW-EXO is composed of 3-D printed polylactic acid parts, aluminum connectors, and a Dynamixel actuator (XM430-W350-R) (see Fig. 6).

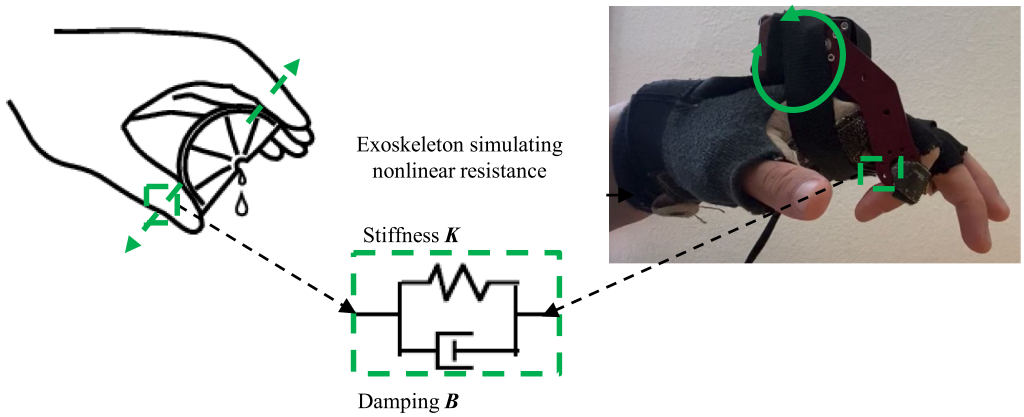

Fig. 5. Exoskeletons used for simulating nonlinear resistances in physical manipulations.

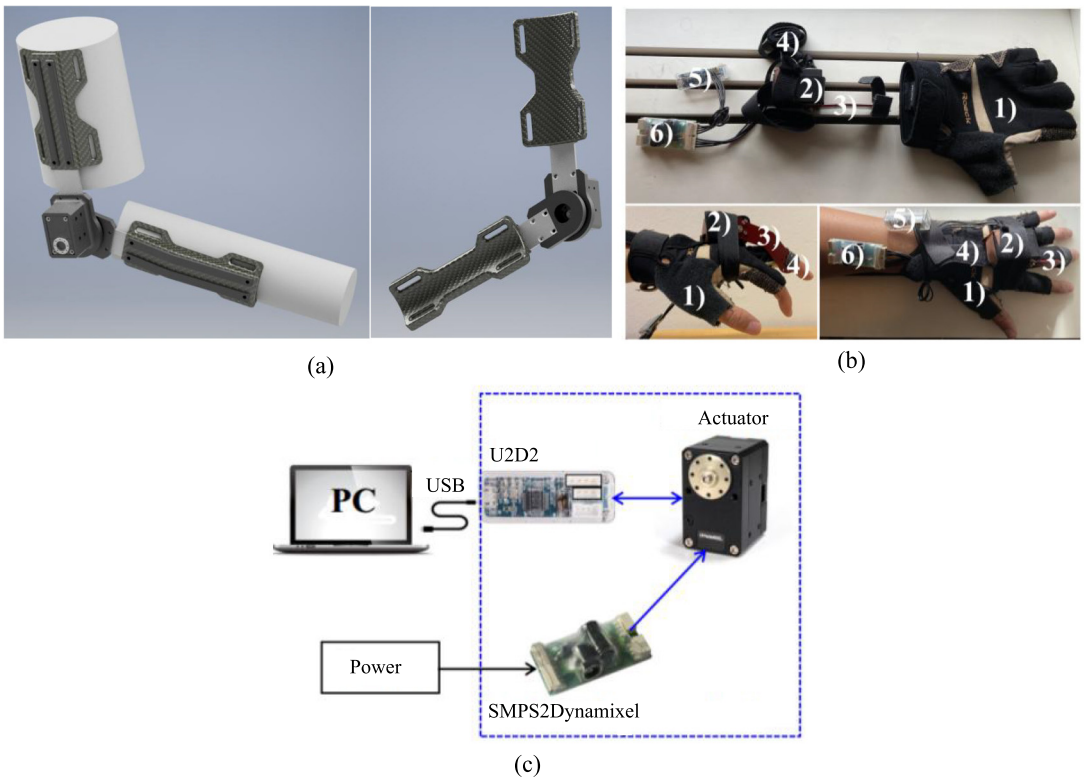

Fig. 6. Elbow and finger exoskeleton designs. (a) POW-EXO. (b) TIE-EXO. (1) Glove. (2) Dynamixel actuator. (3) Rigid shaft. (4) Soft strip bands. (5) USB communication interface. (6) Power converter. (c) Their electrical components.

The desired position $\boldsymbol{q}_d$ and $\dot{\boldsymbol{q}}_d$ of the OIAC were constantly set to initial angle q_0 and 0 (see Eq. (1)),

$$\boldsymbol{q}_d = q_0, \dot{\boldsymbol{q}}_d = 0.0 \tag{14}$$

Exoskeleton wearers[3] were asked to flex their elbows and fingers (as they can) when being subject to the resistant torque $\boldsymbol{\tau}_{fb}$ (see Eq. (1)). This resistance is determined and quantified by the online-tuned impedance parameters $\boldsymbol{K}(t)$ and $\boldsymbol{B}(t)$ (see Eq. (7)). The more they flex, the greater resistance they suffer. Interestingly, this quantification reveals that there are impedance variants and average impedance invariants in human finger and elbow periodic movements. It means that the impedance parameters $\boldsymbol{K}(t)$ and $\boldsymbol{B}(t)$ can online vary, but their averages $\overline{\boldsymbol{K}(t)}$ and $\overline{\boldsymbol{B}(t)}$ converge to specific ranges. These invariant impedance averages can serve as "fingerprints" for identifying various human joint movements in applications (e.g., muscle fatigue) [26]. The experimental video link can be seen at footnote.[4] The nonlinear functions of average position $\overline{q}$, velocity $\overline{\dot{q}}$, and mechanical impedance are given by (see Fig. 8),

$$\overline{q} = f_1(\overline{\boldsymbol{K}}) \approx 0.069 + 0.53\overline{\boldsymbol{K}}^{\frac{1}{2}}, \overline{\dot{q}} = f_2(\overline{\boldsymbol{B}}) \approx 0.3 + 0.65\overline{\boldsymbol{B}}^{\frac{1}{4}} \tag{15}$$

4.3. Multifunctional exoskeleton control

Most existing upper limb exoskeletons are multi-jointed ($\geqslant 2$), heavy (> 1 kg), expensive ($\geqslant 1000$ USD), and single-functional

[3] The wearers consented to the participation in the experiments, which have been approved by the Research Ethics Committee of the University of Southern Denmark (case no.: 20/70421).

[4] https://www.youtube.com/watch?v=GKlrU0ub6R8

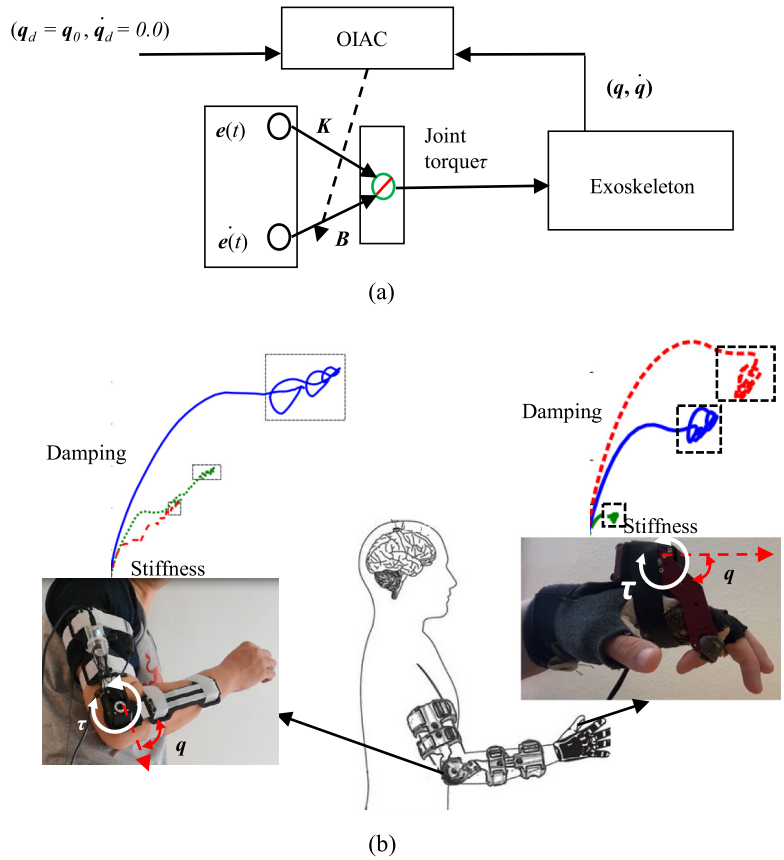

Fig. 7. OIAC control for human finger and elbow mechanical (interactive) impedance quantification. (a) OIAC receives the exoskeleton position q and velocity $\dot{q}$ feedback when producing the joint torque τ, in which its stiffness $\boldsymbol{K}$ and damping $\boldsymbol{B}$ parameters are online tuned (see Eq. (7)). The desired position $\boldsymbol{q}_d$ and $\dot{\boldsymbol{q}}_d$ of the OIAC were constantly set to initial angle q_0 and 0 (see Eq. (1)). (b) The average stiffness $\overline{\boldsymbol{K}(t)}$ and damping $\overline{\boldsymbol{B}(t)}$ were used to quantify the mechanical (interactive) impedance of human fingers and elbows.

Algorithm 1: Optimization of the whip task

Inputs: L_d = 0.10m (threshold distance);
Outputs: Optimal $(q_{i,s},\ q_{i,e},\ q_{f,s},\ q_{f,e},\ D)$ (Eq. (12));
Optimization initialization, $t_0 = 0.05$ [s], $T = 1.2$ [s];
for *DIRECT-L, i = 1 to 600* **do**
  Simulation initialization;
  Update $(q_{i,s},\ q_{i,e},\ q_{f,s},\ q_{f,e},\ D)$ (Eq. (12));
  while $0 \le t < T$ **do**
    Compute $\boldsymbol{\tau}_G$ (Eq. (10));
    while $t_0 \le t < t_0 + D$ **do**
      Compute $\boldsymbol{q}_d$, $\dot{\boldsymbol{q}}_d$, $\boldsymbol{e}$, $\dot{\boldsymbol{e}}$ (Eq. (12) and (6));
      if *OIAC == True* **then**
        Update $\boldsymbol{K}$ and $\boldsymbol{B}$ (Eq. (7));
      else
        Use constant $\boldsymbol{K}$ and $\boldsymbol{B}$ (Eq. (13));
      end
    end
    Compute/command torque inputs $\boldsymbol{\tau}$ (Eq. (1));
    Compute and save distance L;
  end
  Save $L^* = \boldsymbol{Min}$(L);
  if $\boldsymbol{Min}(L^*) < L_d$ **then**
    break;
  end
end
Save the performance $\boldsymbol{Min}(L^*)$ and movement parameters $[q_{i,s}, q_{i,e}, q_{f,s}, q_{f,e}, D]^T$;

(i.e., assist-as-needed (AAN)) [28]. Besides, they are over-dependent on force and muscle sensing, as well as physical compliant mechanisms (e.g., springs). These drawbacks prevent their large-scaled rehabilitation and fitness applications at homes [29,30] (see Fig. 9).

To address it, we proposed a learning-based model for multifunctional elbow exoskeleton control, i.e., assist- and resist-as-needed (AAN and RAN) [31]. The inverse dynamics model of the POW-EXO and its wearer (see Fig. 10) is given by [32],

$$I\ddot{q} + h(\dot{q}) + g(q) = \tau + \tau_h = \tau_s \tag{16}$$

where I, h, and g represent the unknown inertial, viscous, and gravity terms determined by exoskeleton joint acceleration $\ddot{q}$, velocity $\dot{q}$, and position q. The control input τ is generated by the learning-based model (see Fig. 10), compensating for the unknown exoskeleton dynamics (e.g., g) and interaction torque τ_h. The AAN is required to assist patients to perform physical rehabilitation for restoring motor capabilities. In fitness and late rehabilitation, the RAN can produce variable compliant resistances against wearers' movements. The two functions were implemented on a lightweight (0.425 kg) and portable elbow exoskeleton (i.e., POW-EXO) worn by human subjects.

4.3.1. AAN

The AAN control is achieved by combining iterative learning and OIAC mechanisms for generating the torque τ,

$$\tau = -(\tau_{fb} + \tau_{ff}) \tag{17}$$

where τ_{fb} denotes the feedback term (online) adapted by the OIAC at each iteration (see Eqs. (1) and (7)), while τ_{ff} is the feedforward

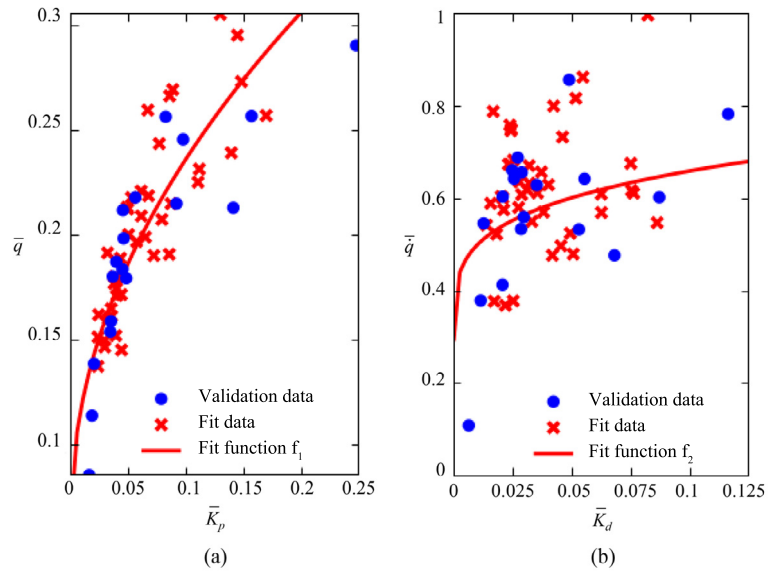

Fig. 8. Nonlinear resistance functions $\bar{q} = f_1(\overline{\boldsymbol{K}(t)})$ and $\bar{q} = f_2(\overline{\boldsymbol{B}})$ (see Eq. (15)).

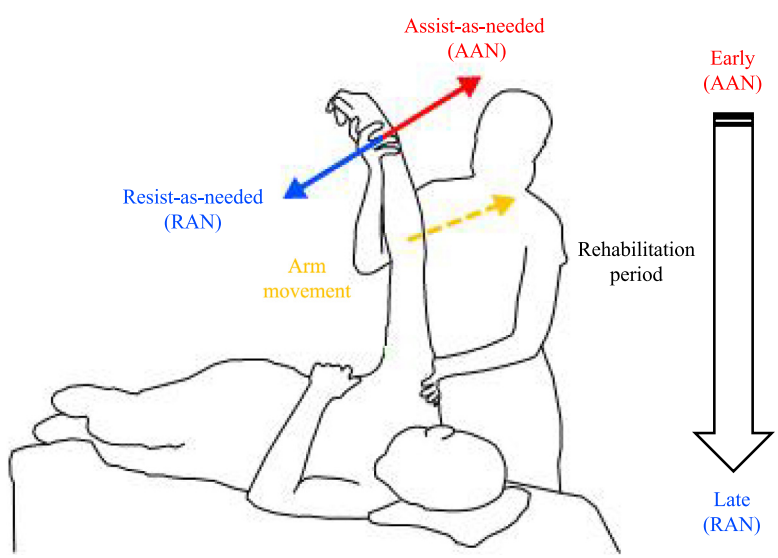

Fig. 9. Assist- and resist-as-needed (AAN and RAN) control in different rehabilitation stages [29,30].

term learned by an iterative mechanism,

$$\tau_{ff}^{i}(t) = \tau_{ff}^{i-1}(t) + \alpha e^{i-1}(t) \tag{18}$$

where the feedforward term τ_{ff} at time t of ith iteration (trial) is determined by itself and position error e at the previous iteration (see e in Eq. (1)). The learning gain α is empirically set to 0.1 for achieving fast stable control. The desired exoskeleton joint position q_d is given by,

$$q_d(t) = -\frac{\pi}{2.0}|\sin(\frac{4\pi}{15.0}t)|[rad],\ t \in [0,\ 15.0][s] \tag{19}$$

where 15.0 is the task time. Such desired position prevents elbow's over-extension (i.e., $q_d(t) > 0$), thereby guaranteeing wearer's safety. In AAN control, parameter θ is learned to minimize the position error given by,

$$\begin{aligned} &\text{minimize} && |q_d - q(\theta)|,\ \theta = [\tau_{ff}, K_p, K_d]^T \\ &\text{subject to} && I\ddot{q} + h(\dot{q}) + g(q) = \tau_s(\theta), \\ & && \tau_s(\theta) = \tau(\theta) + \tau_h,\ \tau(\theta) = -(\tau_{ff} + \tau_{fb}) \end{aligned} \tag{20}$$

where τ_s denotes the sensed torque of the POW-EXO actuator. The iterative learning and online impedance adaptation mecha-

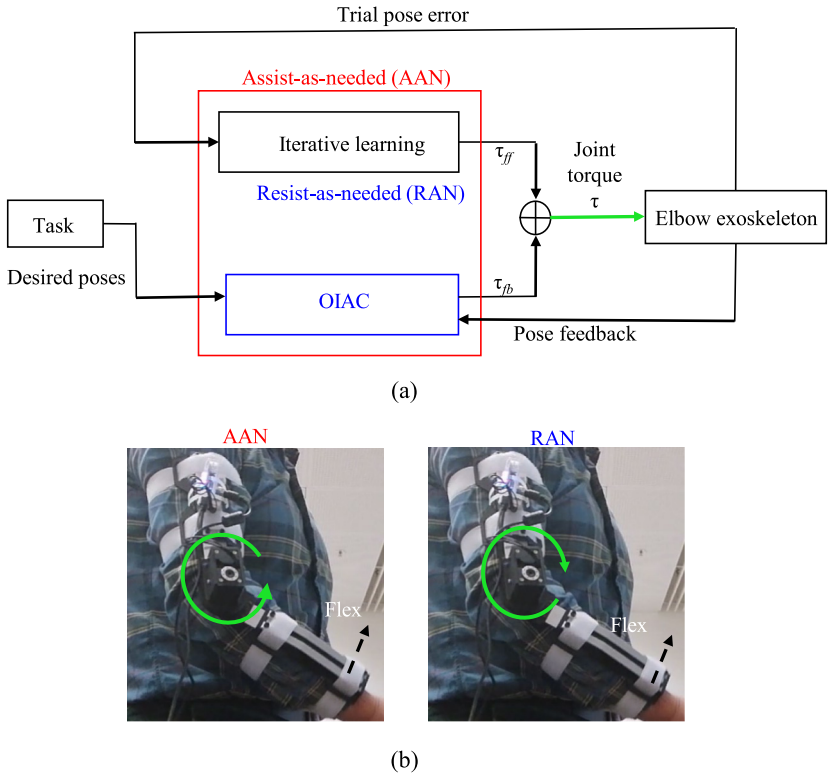

Fig. 10. Multifunctional exoskeleton control. (a) Integration of the OIAC and iterative learning. (b) Assist-as-needed (AAN) and resist-as-needed (RAN) functions when flexing elbows.

nisms are integrated for learning predictive (i.e., τ_{ff}) and adaptive (i.e., (K_p, K_d)) terms. Note that the AAN here enables the POW-EXO to generate motion assistance for compensating for involuntary (i.e., reflex) movement by various wearers. This is different from conventional AAN control that aims at maximizing wearers' involvement. The AAN result of the proposed iterative learning and OIAC can be seen in Fig. 11.

4.3.2. Resist-as-needed (RAN) control

Unlike the proposed AAN, only the adaptive terms K_p and K_d of the OIAM are online tuned for various exercisers. Therefore, the resistant torque is given by (rewritten from Eq. (17)),

$$\tau = -\tau_{fb} \tag{21}$$

This is because its desired exoskeleton joint position is fixed as,

$$q_d(t) = 0.0 \tag{22}$$

In contrast to the ANN, there is no cyclic tracking task. Moreover, the OIAM acts only as a virtual adjustable spring and damper for exercises. Greater resistant torque (i.e., τ) results from larger joint position (i.e., $|e|$) and velocity (i.e., $|\dot{e}|$) errors (see Eq. (7)). In RAN control, parameter Ω is online modulated to maximize wearer's exercise involvement quantified by,

$$\begin{aligned} &\text{maximize} && |\tau_s(\Omega)|,\ \Omega = [K_p, K_d]^T \\ &\text{subject to} && I\ddot{q} + h(\dot{q}) + g(q) = \tau_s(\Omega), \\ & && \tau_s(\Omega) = \tau(\Omega) + \tau_h,\ \tau(\Omega) = -\tau_{fb}(\Omega) \end{aligned} \tag{23}$$

the RAN result of the proposed OIAC can be seen in Fig. 12.

The above implementation relied only on internal pose (e.g., joint position) feedback, rather than physical compliant mechanisms (e.g., springs) and external sensors (e.g., electromyography (EMG) or force) typically required by conventional exoskeletons and controllers [33]. The proposed model provides a novel technique to achieve multifunctional exoskeleton control with minimal mechatronics and sensing. Interestingly, its RAN control and POW-EXO as a quantification means, may reveal interactive (mechanical) impedance variants and invariants in human elbow control (see Fig. 7). The experimental video link can be seen at footnote.[5]

4.4. Robot-inspired human arm control hypothesis

Human arm motor control principles can be abstracted and utilized to enhance robot counterpart performance in biorobotics [5,6]. Very little attention has been paid in a reverse way, i.e., how robot control contributes to research on neuromechanical control of human arm swings remains unclear. For instance, it is no doubt that human shoulders require more control efforts than those at their elbows when periodically swinging arms, since the shoulders are closer to body trunks. What can be a control effort percentage spent at a shoulder joint? Does an elbow joint need active control from central (e.g., neural) control? Answering these questions will facilitate our understanding of human arm control and assist design complexity reductions of wearable robots.

To answer the questions, we proposed a fast ($<$ 10 trials) sensorimotor learning and adaptation (SEED) model on a two-jointed robot to raise a novel hypothesis for human arm control (see Fig. 13 (a)) [34]. An inverse dynamics model of the two-joint robot arm is given by,

$$M(q)\ddot{q} + H(q, \dot{q})\dot{q} + G(q) = \tau_{in} + \tau \tag{24}$$

[5] https://www.youtube.com/watch?v=oFDWYAt9tz8

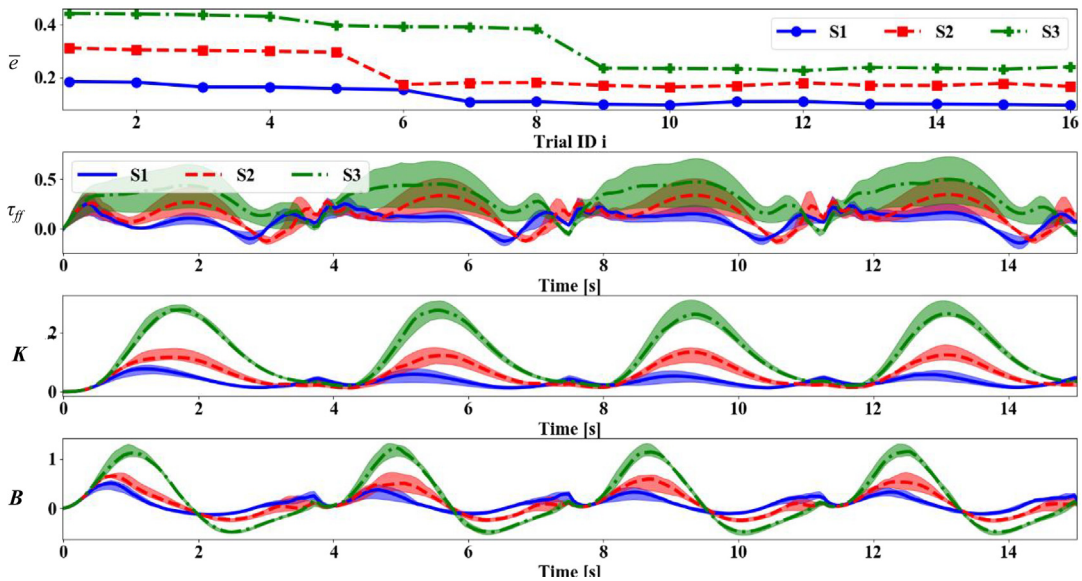

Fig. 11. Statistical results of the proposed model for AAN tasks. In the tasks, three subjects (i.e., S1–S3) wore the POW-EXO, and 16 trials were run for each subject. Rows 1–4 illustrate the average position error $\bar{e}$ [*rad*], the medians and deviations of learned feedforward torque τ_{ff}, and impedance parameters $\boldsymbol{K}(t)$ and $\boldsymbol{B}(t)$. Blue, red, and green colors depict the results of subjects S1–S3, respectively.

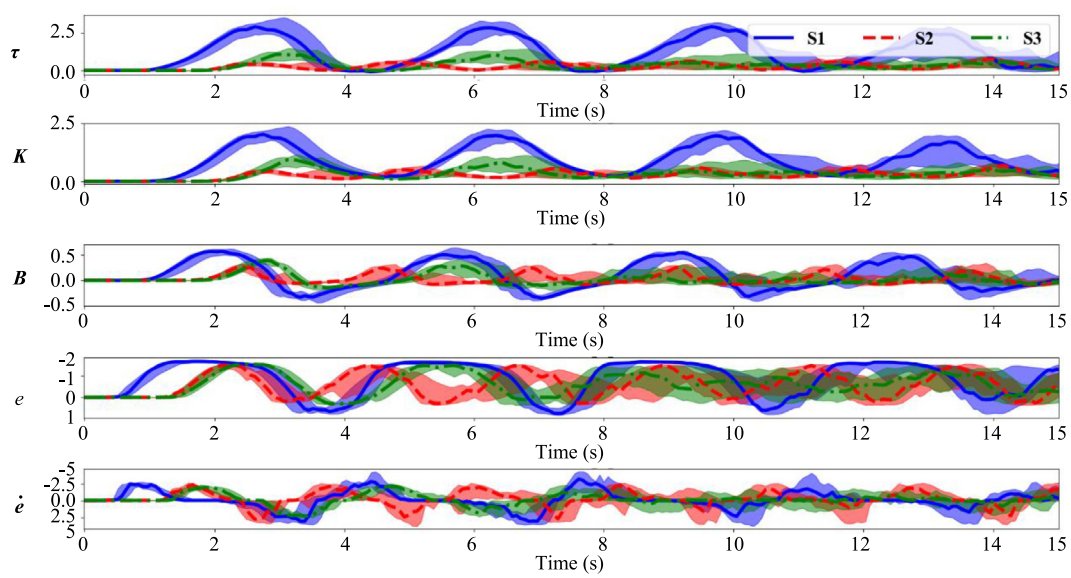

Fig. 12. Statistical results of the proposed online impedance adaptation control for RAN tasks. In the tasks, three subjects (i.e., S1–S3) wore the POW-EXO, and 16 trials were run for each subject. Rows 1–5 illustrate the medians and deviations of the joint control torque τ (see Eq. (21)), impedance parameters $\boldsymbol{K}(t)$ and $\boldsymbol{B}(t)$, the position e and velocity $\dot{e}$ errors. Blue, red, and green colors depict the results of subjects S1–S3, respectively.

where q, $\dot{q}$, $\ddot{q} \in \mathbb{R}^2$ denote the position, velocity, and acceleration of the two joints of the arm. $M(q) \in \mathbb{R}^{2\times 2}$, $H(q,\dot{q}) \in \mathbb{R}^{2\times 2}$, and $G(q) \in \mathbb{R}^2$ are unknown inertial, Coriolis/centrifugal, and gravitational matrices. The torques $\tau_{in} \in \mathbb{R}^2$ result from interaction between joints, while the input (i.e., control) torque τ is generated by the online SEED model. Note that all arm dynamic parameters (e.g., $M(q)$) in Eq. (24) are unknown for all implementations presented in this paper.

The SEED model consists of a Gaussian motor primitive (GATE) neural network (as feedforward control) and the OIAC (see Fig. 13(b)). Therefore, the control torque τ is given by:

$$\tau = \tau_{ff} + \tau_{fb} \tag{25}$$

where $\tau_{ff} \in \mathbb{R}^2$ and $\tau_{fb} \in \mathbb{R}^2$ denote the feedforward and feedback torques produced by a feedforward Gaussian motor primitive (GATE) neural network and feedback OIAC. Sensorimotor weights over trials are learned in the GATE network, while the OIAC is used to online tune impedance gains in a trial. The network has input, middle, and output layers of neurons organized in a feedforward and fully connected way (see Fig. 14). The seven input neurons are given by:

$$\theta_t^{(i+1)} = [1, q_0, \dot{q}_0, \ddot{q}_0, q_1, \dot{q}_1, \ddot{q}_1]^T \in \mathbb{R}^{7\times 1} \tag{26}$$

where $q_{(0,1)}$, $\dot{q}_{(0,1)}$, $\ddot{q}_{(0,1)}$ are the joint positions, velocities, and accelerations of the two-joint robot arm (see Fig. 13) at the time t of the $(i+1)$th trial. The input neurons $\theta_t^{(i+1)}$ are weightedly transmitted to 20 middle neurons,

$$\Omega^{(i+1)} = \begin{bmatrix} \omega_{0,0} & \omega_{0,1} & \cdots & \omega_{0,18} & \omega_{0,19} \\ \omega_{1,0} & \omega_{1,1} & \cdots & \omega_{1,18} & \omega_{1,19} \end{bmatrix} \in \mathbb{R}^{14\times 20} \tag{27}$$

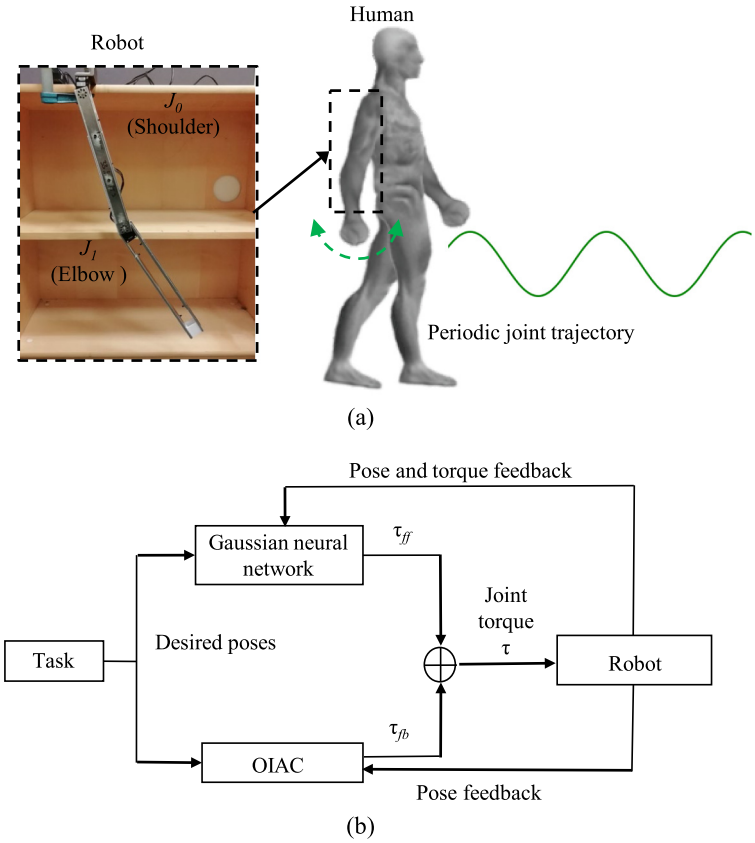

Fig. 13. From robot to biomechanics in arm swing control. (a) Control effort gradient from a shoulder (97%) to elbow (3%). (b) Integration of the OIAC into a Gaussian neural network.

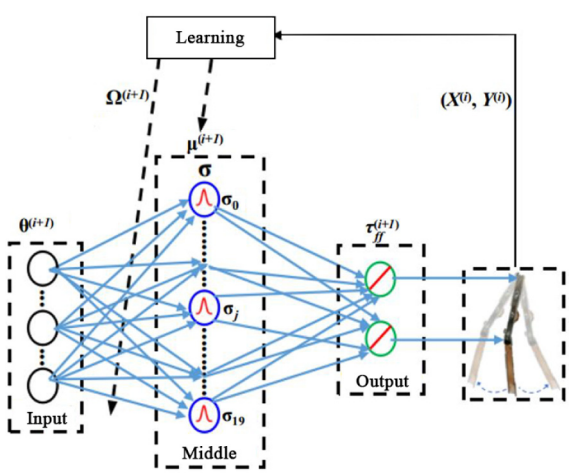

Fig. 14. Feedforward learning control based on a Gaussian motor primitive (GATE) neural network. It learns to generate the feedforward joint torques $\tau_{ff,s}^{(i+1)}$ $(s = 0, 1)$ during trials.

where $\omega_{s,j} \in \mathbb{R}^{7\times 1}$ denotes the weights of the input neurons $\theta_t^{(i+1)}$ with respect to the sth joint and the jth Gaussian motor primitive (GATE). Each GATE neuron is given by a multivariate Gaussian function:

$$\sigma_j(\theta_t^{(i+1)}, \mu_j^{(i+1)}) \sim N(\mu_j^{(i+1)}, \Sigma_j^{(i+1)}), j = 0, 1, \ldots, 18, 19 \tag{28}$$

where σ_j denotes the jth GATE neuron output with respect to the input neurons $\theta_t^{(i+1)}$. The number of the GATE neurons is set to 20 for stable and fast learning track control. The two output neurons are the feedforward torques $\tau_{ff,s}^{(i+1)}$ $(s = 0, 1)$ at the time t of the $(i+1)$th trial,

$$\tau_{ff,s}^{(i+1)}(\theta_t^{(i+1)}) = \frac{\sum_{j=0}^{j=19}(\sigma_j(\theta_t^{(i+1)}, \mu_j^{(i+1)})(\theta_t^{(i+1)})^T \omega_{s,j}^{(i+1)})}{\sum_{j=0}^{j=19}(\sigma_j(\theta_t^{(i+1)}, \mu_j^{(i+1)}))} \tag{29}$$

where the feedforward torques are the normalized (i.e., $\sigma_j(\theta_t^{(i+1)}, \mu_j^{(i)})$) and weighted (i.e., $\omega_{s,j}^{(i+1)}$) sum of the GATE and input neurons. Due to its online learning, the proposed GATE feedforward learning is more computationally efficient, compared to classical regression models (e.g., RBF or LWPR) characterized by the incremental learning [35,36]. The mean vector $\mu_j^{(i+1)}$ of the jth GATE neuron in the $(i+1)$th trial is given by:

$$\mu_j^{(i+1)} = (\overline{X}_{(j\Delta t \to (j+1)\Delta t)}^{(i)})^T \in \mathbb{R}^{7\times 1} \tag{30}$$

where Δt denotes the mean row length of the trained input matrix $X^{(i)}$ during t time steps in the ith trial,

$$X^{(i)} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{(t-2)} \\ x_{(t-1)} \end{bmatrix} \in \mathbb{R}^{t\times 7}, x_t = (\theta_t^{(i)})^T \tag{31}$$

where X is the trained (sensor) input matrix, in which each row x_t consists of joint positions, velocities, and accelerations (see Eq. (26)) at the time step t. The weights $\omega_{s,j}^{(i+1)}$ (see Eq. (29)) in the $(i+1)$th trial is updated by:

$$\omega_{s,j}^{(i+1)} = ((X^{(i)})^T \delta_j^{(i+1)} X^{(i)})^{-1}((X^{(i)})^T \delta_j^{(i+1)} Y_s^{(i)}), s = 0, 1 \tag{32}$$

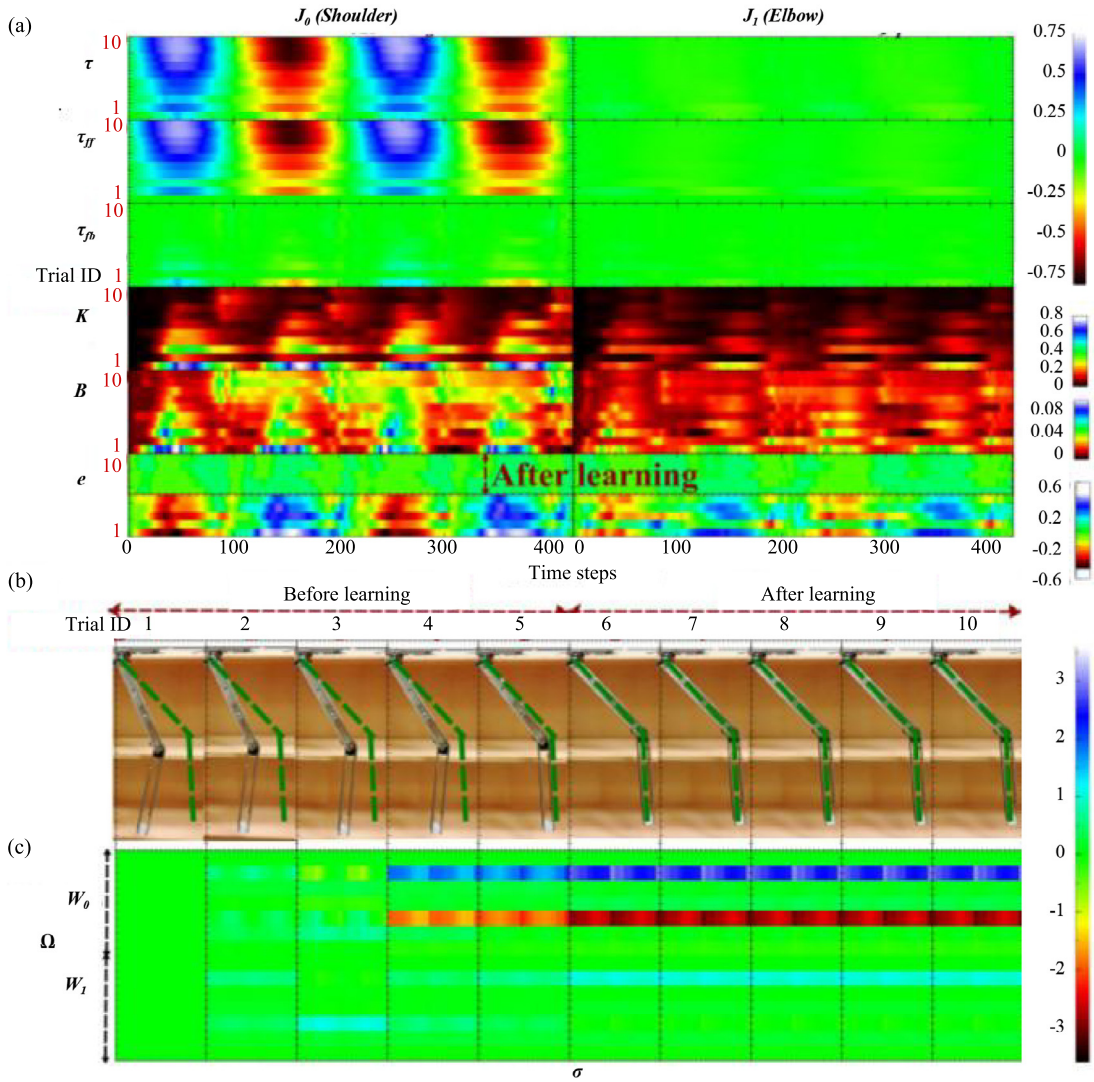

Fig. 15. Periodic joint trajectory tracking undertaken by the online SEED model during 10 trials. The learning and adaptation applies to the two-joint arm joints J_0 and J_1 (see Fig. 13). (A) Overall torques τ consist of the feedforward τ_{ff} and feedback τ_{fb} torques. τ_{ff} is learned by the GATE neural network (see Fig. 14 and Eq. (32)), while τ_{fb} are determined by impedance gains $\boldsymbol{K}(t)$ and $\boldsymbol{b}(t)$ tuned by the OIAC. (B) Snapshots during the learning trials. (C) Sensorimotor weights Ω learned during 10 trials ($W_i = \omega_{i,0-19} \in \mathbb{R}^{7\times 20}$, $i = 0, 1$, see Eq.((27))). Its learning is based on the 20 GATE neurons σ (see Fig. 14).

where $\delta_j^{(i+1)} \in \mathbb{R}^{t\times t}$ denotes the weighted diagonal matrix based on the jth GATE neuron (see Eq. (28)),

$$\begin{aligned}\delta_j^{(i+1)} =& diag(\sigma_j(x_0, \mu_j^{(i+1)}), \sigma_j(x_1, \mu_j^{(i+1)}), \ldots, \\ & \sigma_j(x_{(t-2)}, \mu_j^{(i+1)}), \sigma_j(x_{(t-1)}, \mu_j^{(i+1)}))\end{aligned} \tag{33}$$

and $Y_s^{(i)}$ the trained torque matrix with respect to the sth joint given by,

$$Y_s^{(i)} = [\tau_0, \tau_1, \ldots, \tau_{(t-2)}, \tau_{(t-2)}]^T \in \mathbb{R}^{t\times 1} \tag{34}$$

Updating the neural weights $\omega_{s,j}^{(i+1)}$ in the $(i+1)$th trial minimizes the error $J(\omega_{s,j}^{(i+1)})$ of the weighted regression based on the trained input $X^{(i)}$ and output $Y_s^{(i)}$ matrices,

$$J(\omega_{s,j}^{(i+1)}) = \frac{1}{2}(X^{(i)}\omega_{s,j}^{(i+1)} - Y_s^{(i)})^T \delta_j^{(i)}(X^{(i)}\omega_{s,j}^{(i+1)} - Y_s^{(i)}) \tag{35}$$

The minimization and derivation of Eqs. (35) and (32) can be seen in [34]. All parameters and their determinations of the GATE neural network are given in Table B.4 (see Fig. 15).

The model was used to control periodic arm swing in a two-joint robot arm. As a result, the proposed model enabled the arm to stably learn the tasks within 10 trials (see Fig. 15), compared to thousands of trials required by state-of-art learning algorithms (see Table 3). This model facilitates the exploration of unknown human arm dynamics, in which the elbow joint requires much less active control compared to the shoulder. This control percentage goes below to 3% of the overall effort, while the shoulder one is 97%. The joint control effort is quantified by the joint control power consumption ratios PC_s,

$$PC_s = \frac{P_s}{\sum_{s=0}^{s=1} P_s}, P_s = \frac{\sum_{t=1}^{t=m} VI(t)}{m}, s = 0 \text{ (shoulder)}, 1 \text{ (elbow)} \tag{36}$$

where P_s denotes the average control power consumption of the joint J_s during time steps m. $V = 12.0V$ is the Dynamixel (XM430-W350-R) motor voltage, while $I(t)$ denotes the motor current given by internal sensors.

Note that the online SEED is a trial-based learning model starting at the proposed OIAC feedback controller, in which the impedance parameters $\boldsymbol{K}(t)$ and $\boldsymbol{B}(t)$ are online tuned to track the desired joint positions q_d and velocities $\dot{q}_d$.

Table 3
Comparison between our proposed (SEED) and state-of-art methods.

Algorithm	SEED	EEL [37]	ILC [38]	GANs [39]	DRL [40]
Year	2023	2022	2021	2020	2020
Predictive learning	Yes	Yes	Yes	Yes	Yes
Online impedance adaptation	Yes	No	No	No	Yes
Physics-based	Yes	Yes	Yes	No	No
Data-driven	Yes	Yes	No	Yes	Yes
Online learning and adaptation	Yes	No	No	No	No
Dynamics exploitation	Yes	No	No	No	No
Real robot	Yes	Yes	Yes	No	Yes
Incremental learning	No	Yes	No	Yes	Yes
Trials	< 10	> 400	14	100	>7000

Algorithm 2: Online SEED

Inputs: $X^{(i)}$ and $Y_s^{(i)}$ (Eqs. (31) and (34)); $q^{(i+1)}$ and $q_d^{(i+1)}$ (Eq. (1)) $(i \geq 1)$;
Outputs: $\tau^{(i+1)}$, $\tau_{ff}^{(i+1)}$, and $\tau_{fb}^{(i+1)}$ (Eq. (25)) ;
Learned: $\tau_{ff}^{(i+1)}$, $\boldsymbol{K}^{(i+1)}(t)$, and $\boldsymbol{B}^{(i+1)}(t)$ (Eq. (7)) ;
while ***trial*** $i < i_{max}$ **do**
  Obtain the trained data $X^{(i)}$ and $Y_s^{(i)}$ in the ith trial;
  Update the weights $\omega_{s,j}^{(i+1)}$ (Eq. (32)) ;
  Learn the feedforward torque $\tau_{ff}^{(i+1)}$ in the $(i+1)$th trial (Eq. (29)) ;
  while ***time*** $t < t_{max}$ **do**
    Adapt the impedance parameters $\boldsymbol{K}^{(i+1)}(t)$ and $\boldsymbol{B}^{(i+1)}(t)$ (Eq. (7)) ;
    Compute the feedback torques $\tau_{fb}^{(i+1)}(t)$ (Eq. (1)) ;
    Compute the control torques $\tau^{(i+1)}(t) = \tau_{ff}^{(i+1)}(t) + \tau_{fb}^{(i+1)}(t)$;
  end
  Save the trained data $X^{(i+1)}$ and $Y_s^{(i+1)}$ in the $(i+1)$th trial;
end

This novel hypothesis can be raised for future validations in real human arm control, i.e., shoulder joints are actively controlled while elbows are actuated passively by interaction torques between shoulders and elbows. It shows how a neural network leverages joint mechanics to simplify arm movement control. The proposed SEED outperforms pure data-driven methods such as deep reinforcement learning, end-to-end neural learning and generative adversarial networks (GANs) in fewer trials [37,39]. For instance, Bogdanovic et al. applied Deep Deterministic Policy Gradient (DDPG) to train online impedance control of a two-jointed hopping robot [40]. This train requires more 7000 trials performed in simulation. To reduce training trials, Zhang et al. developed a neural network based iterative learning control (ILC) for one-joint control of a SCARA robot [38]. Although this physics-based learning only requires 14 trials, it does not exhibit faster learning (< 10 trials), inverse dynamics exploitation and adaptation. Data-driven learning and physics-based adaptation are integrated to achieve these in the proposed SEED. It shows that online physics-based (impedance) adaptation can boost data-driven learning in inverse arm dynamics control. It exhibits fast learning and gradient exploitation of joint dynamics, which have not yet been realized by the state-of-art fast learning methods such as feedforward neural networks and iterative learning control. Taken together, the proposed SEED model paves a way for implementing data-efficient sensorimotor learning and adaptation of human-like arm movement. The experimental video link can be seen at footnote.[6] The multi-joint control gradient hypothesis will be validated in 10 healthy human subjects by wireless EMG (Trigno Avanti) sensors and a 3D motion (optitrack) capture system in future (see Table 3).

5. Conclusion

Mechanical impedance is an important property to characterize and regulate the robot motion control and the human motor control, and to describe and harmonize interactive behaviors between humans, robots and external environments. Variable impedance control is ubiquitous and indispensable in these complicated motions and behaviors of humans and robots. In this tutorial, an implementation of the variable impedance control OIAC, was introduced to demonstrate its applications in different scenarios including human-like rapid manipulation of a flexible whip, interpretation of human motor control, and human impedance quantification and robot impedance adaptation in exoskeleton control as an example of physical human–robot collaboration. With the shared source codes for OIAC, this tutorial is expected to enrich the toolbox of the interactive robotics and broaden the applications of variable impedance control, and to facilitate the research in robot control and human motor control. In the future, the OIAC will be extended to apply to admittance control of leg joint control [41]. A tutorial proposal will be submitted to the IEEE flagship robotics conference IROS (2023) for increasing its impacts and usages.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank Cao Danh Do for robotic arm design. This work was supported by the Human Frontier Science Program (RGP0002/2017), the Brødrene Hartmanns Fund (A36775), and the Thomas B. Thriges Fund (7648-2106).

Appendix A. Tutorial code

The Python code (on the OIAC) can be seen at https://github.com/lenonrobot/Online-Impedance-Adaptation-Control/blob/main/oiac.py.

Appendix B. GATE neural network parameters

All parameters and their determinations of the GATE neural network are given in Table B.4.

[6] https://www.youtube.com/watch?v=e6UD3aYlJaw

Table B.4
Parameters and their determinations of GATE neural network.

Parameter	Denotation	Determination
θ	Input vector consisting of joint positions, velocities, and accelerations, Eq. (26)	Joint feedback (from motor encoders)
Ω	Learned weight matrix, Eq. (27)	Minimization based on non-linear regression, Eq. (32)
μ	Mean vector of Gaussian motor primitive (GATE), Eq. (28)	Trained input X, Eq. (31)
σ	GATE output vector, Eq. (28)	Input vector θ and learned weight matrix Ω
τ_{ff}	Feedforward torque vector, Eq. (29)	Input vector θ, GATE output vector σ, and learned weight matrix Ω
X	Trained input matrix, Eq. (31)	Joint feedback (from motor encoders)
Y	Trained torque matrix, Eq. (34)	Control torque vector

Appendix C. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.birob.2023.100100.

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